Chapter 12

Electromagnetic Waves in Matter

12-1 Maxwell's Equations in Matter

A. SOMETHING IS MISSING

(1) For fields in the presence of electric charge of density ρ and electric current, that is, charge in motion, of density $\vec{J}$. We have

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \cdots\cdots \text{Faraday's law}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \cdots\cdots \text{Gauss's law}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

(2) The charge density is separated into two parts

$$\rho = \rho_f + \rho_b = \rho_f - \nabla \cdot \vec{P}$$

$$\nabla \cdot \vec{E} = \frac{1}{\epsilon_0}\left(\rho_f - \nabla \cdot \vec{P}\right)$$

$$\Rightarrow \nabla \cdot \left(\epsilon_0 \vec{E} + \vec{P}\right) = \rho_f$$

$$\Rightarrow \nabla \cdot \vec{D} = \rho_f$$

The current density is separated into two parts

$$\vec{J} = \vec{J}_f + \vec{J}_b = \vec{J}_f + \nabla \times \vec{M}$$

$$\nabla \times \vec{B} = \mu_0\left(\vec{J}_f + \nabla \times \vec{M}\right) + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\Rightarrow \nabla \times \left(\vec{B} - \mu_0 \vec{M}\right) = \mu_0 \vec{J}_f + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\Rightarrow \nabla \times \frac{1}{\mu_0}\left(\vec{B} - \mu_0 \vec{M}\right) = \vec{J}_f + \epsilon_0 \frac{\partial}{\partial t}\vec{E}$$

$$\Rightarrow \nabla \times \vec{H} = \vec{J}_f + \epsilon_0 \frac{\partial}{\partial t}\vec{E} \cdots\cdots \text{(a)}$$

(3) From Gauss's law, we have

$$\epsilon_0 \nabla \cdot \vec{E} = \rho$$

From equation (a), we have

$$\nabla \cdot \left(\nabla \times \vec{H}\right) = \nabla \cdot \vec{J}_f + \epsilon_0 \frac{\partial}{\partial t} \nabla \cdot \vec{E} = \nabla \cdot \vec{J}_f + \frac{\partial \rho}{\partial t}$$

Since

$$\rho = \rho_f + \rho_b$$

thus, we obtain

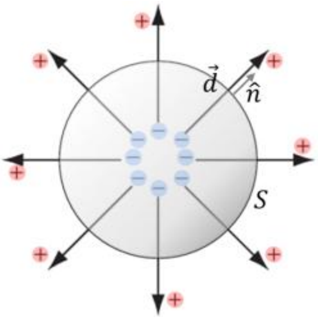
$$\underbrace{\nabla \cdot \left(\nabla \times \vec{H}\right)}_{=0} = \underbrace{\nabla \cdot \vec{J}_f + \frac{\partial \rho_f}{\partial t}}_{=0} + \frac{\partial \rho_b}{\partial t} \neq 0$$

Therefore, an extra current term related to the current density is needed such as

$$\underbrace{\nabla \cdot \vec{J}_P}_{\text{added extra term}} + \frac{\partial \rho_b}{\partial t} = 0$$

B. POLARIZATION CURRENT

(1) Consider a surface $\mathcal{S}$ bounding a volume $\mathcal{V}$ of a nonpolar dielectric. The application of an external field causes the bound charges moving: The positive charges flow out of $\mathcal{V}$ and the negative charges remain within the volume $\mathcal{V}$.



The charge crosses the surface $\mathcal{S}$ is

$$dQ_P = Nq\vec{d} \cdot \hat{n}da = \langle\vec{p}\rangle N \cdot \hat{n}da = \vec{P} \cdot \hat{n}da$$

The net charge through the surface is

$$Q_P = \oint_{\mathcal{S}} \vec{P} \cdot \hat{n}da$$

(2) The bound charge satisfies the charge conservation law:

1. Global conservation of charge

   we started with an electrically neutral dielectric body, the total charge of the body after polarization must remain zero.

   The net charge within the volume is

$$Q_P = -\oint_{\mathcal{S}} \vec{P} \cdot \hat{n}da = \int_{\mathcal{V}} \left(-\nabla \cdot \vec{P}\right) d\tau \text{ and } Q_P = \int_{\mathcal{V}} \rho_b \, d\tau$$

Thus, we obtain

$$\rho_b = -\nabla \cdot \vec{P}$$

The total charge is

$$\oint_{\mathcal{S}} \sigma_b \, da + \int_{\mathcal{V}} \rho_b \, d\tau = \oint_{\mathcal{S}} \frac{dQ_p}{da} da - \int_{\mathcal{V}} \left(\nabla \cdot \vec{P}\right) d\tau$$

$$= \oint_{\mathcal{S}} \vec{P} \cdot \hat{n} \, da - \oint_{\mathcal{S}} \vec{P} \cdot \hat{n} \, da$$

$$= 0$$

2. Local conservation of charge

the charge flows through the surface per unit time is

$$I_P = \frac{dQ_P}{dt} = \oint_{\mathcal{S}} \frac{\partial \vec{P}}{\partial t} \cdot \hat{n} da \text{ and } I_P = \int_{\mathcal{S}} \vec{J}_P \cdot d\vec{a}$$

Thus, we obtain

$$\vec{J}_P = \frac{\partial \vec{P}}{\partial t} \cdots\cdots \text{polarization current}$$

The polarization current satisfy the continuity equation:

$$\frac{\partial \rho_b}{\partial t} + \nabla \cdot \vec{J}_P = \frac{\partial \left(-\nabla \cdot \vec{P}\right)}{\partial t} + \nabla \cdot \frac{\partial \vec{P}}{\partial t} = -\nabla \cdot \frac{\partial \vec{P}}{\partial t} + \nabla \cdot \frac{\partial \vec{P}}{\partial t} = 0$$

(3) Equation (a) becomes

$$\underbrace{\nabla \cdot \left(\nabla \times \vec{H}\right)}_{=0} = \underbrace{\nabla \cdot \vec{J}_f + \frac{\partial \rho_f}{\partial t}}_{=0} + \underbrace{\frac{\partial \rho_b}{\partial t} + \nabla \cdot \vec{J}_P}_{=0}$$

$$= \nabla \cdot \vec{J}_f + \underbrace{\frac{\partial \rho}{\partial t}}_{\rho = \rho_f + \rho_b} + \nabla \cdot \vec{J}_P$$

$$= \nabla \cdot \vec{J}_f + \underbrace{\epsilon_0 \frac{\partial}{\partial t}\left(\nabla \cdot \vec{E}\right)}_{\epsilon_0 \nabla \cdot \vec{E} = \rho} + \nabla \cdot \frac{\partial \vec{P}}{\partial t}$$

$$= \nabla \cdot \left(\vec{J}_f + \epsilon_0 \frac{\partial \vec{E}}{\partial t} + \frac{\partial \vec{P}}{\partial t}\right)$$

$$= \nabla \cdot \left(\vec{J}_f + \frac{\partial}{\partial t}\left(\epsilon_0 \vec{E} + \vec{P}\right)\right)$$

$$= \nabla \cdot \left(\vec{J}_f + \frac{\partial \vec{D}}{\partial t}\right)$$

$$\Rightarrow \nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

C. MAXWELL'S EQUATIONS IN MATTER

(1) For fields in the presence of charge density

$$\rho = \rho_f + \rho_b$$

and electric current density

$$\vec{J} = \vec{J}_f + \vec{J}_b + \vec{J}_P$$

we have

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \cdots\cdots \text{Faraday's law}$$

$$\nabla \cdot \vec{D} = \rho_f \cdots\cdots \text{Gauss's law}$$

$$\nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

(2) For linear media, we have

$$\epsilon = \epsilon_0(1 + \chi_e)$$

$$\mu = \mu_0(1 + \chi_m)$$

and obtain

$$\vec{D} = \epsilon_0\vec{E} + \vec{P} = \epsilon_0\vec{E} + \chi_e\epsilon_0\vec{E} = \epsilon_0(1 + \chi_e)\vec{E} = \epsilon\vec{E}$$

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} = \frac{\vec{B}}{\mu_0} - \chi_m\vec{H} \Rightarrow \vec{H} = \frac{\vec{B}}{\mu_0(1 + \chi_m)} = \frac{\vec{B}}{\mu}$$

Thus, Maxwell's equations become

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \cdots\cdots \text{Faraday's law}$$

$$\nabla \cdot \vec{E} = \frac{\rho_f}{\epsilon} \cdots\cdots \text{Gauss's law}$$

$$\nabla \times \vec{B} = \mu\vec{J}_f + \mu\epsilon\frac{\partial \vec{E}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

D. BOUNDARY CONDITIONS AT THE SURFACE FOR MAXWELL'S EQUATIONS

(1) Maxwell's equations in matter in integral form are

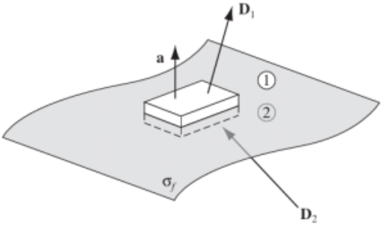
$$\oint_S \vec{D} \cdot d\vec{a} = q_f \cdots\cdots \text{Gauss's law}$$

$$\oint_{\mathcal{S}} \vec{B} \cdot d\vec{a} = 0$$

$$\oint_{\mathcal{C}} \vec{E} \cdot d\vec{s} = -\int_{\mathcal{S}} \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a} \cdots\cdots \text{Faraday's law}$$

$$\oint_{\mathcal{C}} \vec{H} \cdot d\vec{s} = I_f + \int_{\mathcal{S}} \frac{\partial \vec{D}}{\partial t} \cdot d\vec{a}$$

(2) We choose a Gaussian surface for a very tiny area $d\vec{a}$ and let the thickness go to zero.



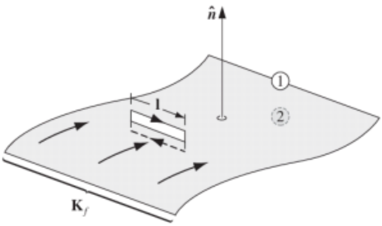
Thus, we obtain

$$\oint_{\mathcal{S}} \vec{D} \cdot d\vec{a} = \sigma_f da \Rightarrow \underbrace{\vec{D}_1 \cdot d\vec{a}}_{\text{media ①}} - \underbrace{\vec{D}_2 \cdot d\vec{a}}_{\text{media ②}} = \sigma_f da \Rightarrow D_{1\perp} - D_{2\perp} = \sigma_f$$

$$\oint_{\mathcal{S}} \vec{B} \cdot d\vec{a} = 0 \Rightarrow \underbrace{\vec{B}_1 \cdot d\vec{a}}_{\text{media ①}} - \underbrace{\vec{B}_2 \cdot d\vec{a}}_{\text{media ②}} = 0 \Rightarrow B_{1\perp} - B_{2\perp} = 0$$

$\Rightarrow B_{\perp}$ is continuous across the surface.

(3) We can choose a closed loop such that the width goes to zero as



Thus, we obtain

$$\oint_{\mathcal{C}} \vec{E} \cdot d\vec{s} = \int_{\mathcal{S}} \left(\nabla \times \vec{E}\right) \cdot d\vec{a} = -\int_{\mathcal{S}} \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a} = 0$$

$$\Rightarrow \underbrace{\vec{E}_1 \cdot d\vec{s}}_{\text{media ①}} - \underbrace{\vec{E}_2 \cdot d\vec{s}}_{\text{media ②}} = 0$$

$$\Rightarrow E_{1\parallel} - E_{2\parallel} = 0$$

$\Rightarrow E_{\parallel}$ is continuous across the surface.

$$\oint_{\mathcal{C}} \vec{H} \cdot d\vec{s} = \int_{\mathcal{S}} \left(\nabla \times \vec{H}\right) \cdot d\vec{a} = I_f$$

$$\Rightarrow \underbrace{\vec{H}_1 \cdot d\vec{s}}_{\text{media ①}} - \underbrace{\vec{H}_2 \cdot d\vec{s}}_{\text{media ②}} = I_f = \vec{K}_f \cdot \left(\hat{n} \times d\vec{s}\right) = \left(\vec{K}_f \times \hat{n}\right) \cdot d\vec{s}$$

$$\Rightarrow H_{1\parallel} - H_{2\parallel} = \vec{K}_f \times \hat{n}$$

(4) For linear media,

$$\vec{D} = \epsilon \vec{E}$$

$$\vec{H} = \frac{\vec{B}}{\mu}$$

we have the boundary conditions as follows

continuous:

$$E_{1\parallel} - E_{2\parallel} = 0$$
$$B_{1\perp} - B_{2\perp} = 0$$

discontinuous:

$$D_{1\perp} - D_{2\perp} = \sigma_f \Rightarrow \epsilon_1 E_{1\perp} - \epsilon_2 E_{2\perp} = \sigma_f$$

$$H_{1\parallel} - H_{2\parallel} = \vec{K}_f \times \hat{n} \Rightarrow \frac{B_{1\parallel}}{\mu_1} - \frac{B_{2\parallel}}{\mu_2} = \vec{K}_f \times \hat{n}$$

12-2 Electromagnetic Waves in Matter

A. ELECTROMAGNETIC WAVES IN LINEAR MEDIA

(1) Suppose that in linear media,

$$\vec{D} = \epsilon \vec{E} \text{ and } \vec{H} = \frac{1}{\mu}\vec{B}$$

there is no free charge or free current. Maxwell's equations become

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \cdots\cdots \text{Faraday's law}$$

$$\nabla \cdot \vec{E} = 0 \cdots\cdots \text{Gauss's law}$$

$$\nabla \times \vec{B} = \mu\epsilon \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

(2) Thus, electromagnetic waves propagate through a linear homogeneous medium at a speed v,

$$\nabla^2 \vec{E} = \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \mu\epsilon \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$\Rightarrow v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{n}$$

where

$$n = \sqrt{\frac{\mu\epsilon}{\mu_0\epsilon_0}}$$

is the index of refraction of the substance. For most material,

$$\chi_m \to 0$$

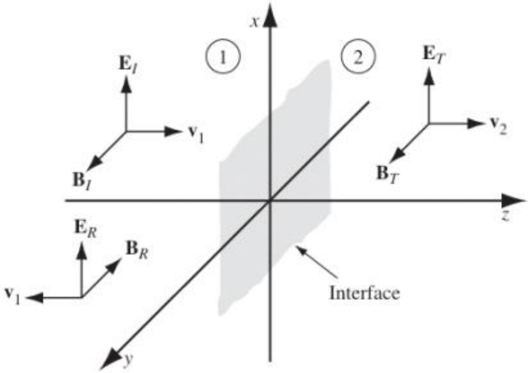
so, we have

$$n = \sqrt{\frac{\mu_0(1+\chi_m)\epsilon}{\mu_0\epsilon_0}} \approx \sqrt{\frac{\epsilon}{\epsilon_0}} = \sqrt{\kappa} > 1$$

Thus, we conclude that light travels more slowly through matter.

B. REFLECTION AND TRANSMISSION AT NORMAL INCIDENCE

(1) Suppose the xy plane forms the boundary between two linear media. A plane wave of frequency ω, traveling in the z direction and polarized in the x direction, approaches the interface from the left:



$$\vec{E}_i(z,t) = E_{0i} e^{i(k_1 z - \omega t)} \hat{x}$$
$$\vec{B}_i(z,t) = E_{0i} e^{i(k_1 z - \omega t)} \frac{1}{v_1} \hat{v}_1 \times \hat{x} = \frac{E_{0i}}{v_1} e^{i(k_1 z - \omega t)} \hat{y}$$

It gives rise to a reflected wave and a transmitted wave,

$$\vec{E}_r(z,t) = E_{0r} e^{i(-k_1 z - \omega t)} \hat{x}$$
$$\vec{B}_r(z,t) = E_{0r} e^{i(-k_1 z - \omega t)} \frac{1}{v_1} \left(-\hat{v}_1\right) \times \hat{x} = -\frac{E_{0r}}{v_1} e^{i(-k_1 z - \omega t)} \hat{y}$$
$$\vec{E}_t(z,t) = E_{0t} e^{i(k_2 z - \omega t)} \hat{x}$$
$$\vec{B}_t(z,t) = E_{0t} e^{i(k_2 z - \omega t)} \frac{1}{v_2} \hat{v}_2 \times \hat{x} = \frac{E_{0t}}{v_2} e^{i(k_2 z - \omega t)} \hat{y}$$

(2) At $z = 0$, the boundary conditions give

$$E_{1\parallel} - E_{2\parallel} = 0 \Rightarrow \underbrace{E_{0i} + E_{0r}}_{=E_{1\parallel}} = \underbrace{E_{0t}}_{=E_{2\parallel}}$$
$$B_{1\perp} - B_{2\perp} = 0$$
$$\epsilon_1 E_{1\perp} - \epsilon_2 E_{2\perp} = 0$$
$$\frac{B_{1\parallel}}{\mu_1} - \frac{B_{2\parallel}}{\mu_2} = 0 \Rightarrow \frac{1}{\mu_1} \underbrace{\left(\frac{E_{0i}}{v_1} - \frac{E_{0r}}{v_1}\right)}_{=B_{1\parallel}} = \frac{1}{\mu_2} \underbrace{\frac{E_{0t}}{v_2}}_{=B_{2\parallel}} \Rightarrow E_{0i} - E_{0r} = \beta E_{0t}$$

where

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2} \approx \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

Thus, we obtain

$$E_{0r} = \frac{1-\beta}{1+\beta} E_{0i}$$
$$E_{0t} = \frac{2}{1+\beta} E_{0i}$$

(3) The reflection coefficient R and the transmission coefficient T

Since

$$I = \frac{1}{2}\epsilon v E_0^2$$

we have

$$R = \frac{I_r}{I_i} = \left(\frac{E_{0r}}{E_{0i}}\right)^2 = \left(\frac{1-\beta}{1+\beta}\right)^2 = \left(\frac{1-\frac{n_2}{n_1}}{1+\frac{n_2}{n_1}}\right)^2 = \left(\frac{n_1-n_2}{n_1+n_2}\right)^2$$
$$T = \frac{I_t}{I_i} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1}\left(\frac{E_{0t}}{E_{0i}}\right)^2 = \left(\frac{2}{1+\beta}\right)^2 = \left(\frac{2}{1+\frac{n_2}{n_1}}\right)^2 = \frac{4n_1 n_2}{(n_1+n_2)^2}$$

12-3 Electromagnetic Waves in Conductors

A. ELECTROMAGNETIC WAVES IN CONDUCTOR

(1) Inside a conductor, according to Ohm's law, the (free) current density in a conductor is proportional to the electric field,

$$\vec{J}_f = \sigma \vec{E}$$

Maxwell's equations for linear media is

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \cdots\cdots \text{Faraday's law}$$

$$\nabla \cdot \vec{E} = \frac{\rho_f}{\epsilon} \cdots\cdots \text{Gauss's law}$$

$$\nabla \times \vec{B} = \mu\sigma\vec{E} + \mu\epsilon\frac{\partial \vec{E}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

(2) The continuity equation for free charge is

$$\nabla \cdot \vec{J}_f + \frac{\partial \rho_f}{\partial t} = 0$$

together with Ohm's law and Gauss's law, gives

$$\frac{\partial \rho_f}{\partial t} = -\nabla \cdot \vec{J}_f = -\nabla \cdot \sigma\vec{E} = -\frac{\sigma}{\epsilon}\rho_f$$

$$\Rightarrow \rho_f(t) = e^{-(\sigma/\epsilon)t}\rho_f(0)$$

Thus, any initial free charge $\rho_f(0)$ dissipates in a characteristic time $\tau \equiv \epsilon/\sigma$.

(3) As accumulated free charge disappears, from then on, $\rho_f = 0$, we have

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \cdots\cdots \text{Faraday's law}$$

$$\nabla \cdot \vec{E} = 0 \cdots\cdots \text{Gauss's law}$$

$$\nabla \times \vec{B} = \mu\sigma\vec{E} + \mu\epsilon\frac{\partial \vec{E}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

Applying the curl, we obtain modified wave equations

$$\nabla \times \left(\nabla \times \vec{E}\right) = \nabla \underbrace{\left(\nabla \cdot \vec{E}\right)}_{=0} - \nabla^2 \vec{E} = -\frac{\partial\left(\nabla \times \vec{B}\right)}{\partial t} = -\mu\sigma\frac{\partial \vec{E}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla \times \left(\nabla \times \vec{B}\right) = \nabla \underbrace{\left(\nabla \cdot \vec{B}\right)}_{=0} - \nabla^2 \vec{B} = \mu\sigma \left(\nabla \times \vec{E}\right) + \mu\epsilon \frac{\partial \left(\nabla \times \vec{E}\right)}{\partial t}$$

$$\Rightarrow \nabla^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\Rightarrow \nabla^2 \vec{B} = \mu\sigma \frac{\partial \vec{B}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{B}}{\partial t^2}$$

Assume that

$$\vec{E}(z,t) = E_0 e^{-\kappa z} e^{i(kz-\omega t)} \hat{x}$$

$$\vec{B}(z,t) = \frac{\tilde{k}}{\omega} E_0 e^{-\kappa z} e^{i(kz-\omega t)} \hat{y}$$

we found

$$\tilde{k} = k + i\kappa$$

$$k = \omega \sqrt{\frac{\mu\epsilon}{2}} \left(\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} + 1 \right)^{1/2}$$

$$\kappa = \omega \sqrt{\frac{\mu\epsilon}{2}} \left(\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} - 1 \right)^{1/2}$$

$$\left|\tilde{k}\right| = \sqrt{k^2 + \kappa^2} = \omega \left(\mu\epsilon \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} \right)^{1/2}$$

$$\tan\phi = \frac{\kappa}{k}$$

$$\frac{B_0}{E_0} = \frac{\left|\tilde{k}\right|}{\omega} = \left(\mu\epsilon \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} \right)^{1/2}$$

The electric and magnetic fields are

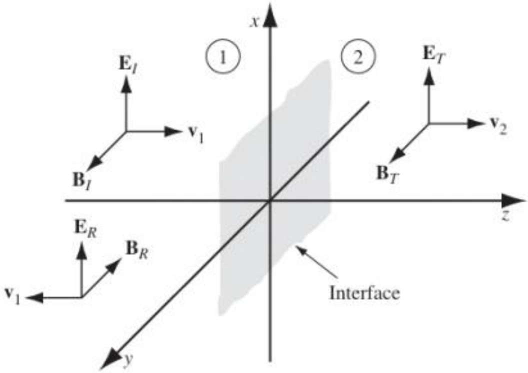
$$\vec{E}(z,t) = E_0 e^{-\kappa z} e^{i(kz-\omega t)} \hat{x}$$

$$\vec{B}(z,t) = \frac{|\tilde{k}|}{\omega} E_0 e^{-\kappa z} e^{i(kz-\omega t+\phi)} \hat{y}$$

where $1/\kappa$ is called the skin depth.

B. REFLECTION AT CONDUCTING SURFACE

(1) Suppose the xy plane forms the boundary between two linear media. A plane wave of frequency ω, traveling in the z direction and polarized in the x direction, approaches the interface from the left:



$$\vec{E}_i(z,t) = E_{0i} e^{i(k_1 z-\omega t)} \hat{x}$$

$$\vec{B}_i(z,t) = \frac{E_{0i}}{v_1} e^{i(k_1 z-\omega t)} \hat{y}$$

It gives rise to a reflected wave and a transmitted wave,

$$\vec{E}_r(z,t) = E_{0r} e^{i(-k_1 z-\omega t)} \hat{x}$$

$$\vec{B}_r(z,t) = -\frac{E_{0r}}{v_1} e^{i(-k_1 z-\omega t)} \hat{y}$$

$$\vec{E}_t(z,t) = E_{0t} e^{i(\tilde{k}_2 z-\omega t)} \hat{x}$$

$$\vec{B}_t(z,t) = \frac{\tilde{k}_2}{\omega} E_{0t} e^{i(\tilde{k}_2 z-\omega t)} \hat{y}$$

(2) At $z = 0$, the boundary conditions give

$$E_{1\parallel} - E_{2\parallel} = 0 \Rightarrow \underbrace{E_{0i} + E_{0r}}_{=E_{1\parallel}} = \underbrace{E_{0t}}_{=E_{2\parallel}}$$

$$B_{1\perp} - B_{2\perp} = 0$$

$$\epsilon_1 E_{1\perp} - \epsilon_2 E_{2\perp} = \sigma_f$$

$$\frac{B_{1\parallel}}{\mu_1} - \frac{B_{2\parallel}}{\mu_2} = \vec{K}_f \times \hat{n} \Rightarrow \frac{1}{\mu_1}\underbrace{\left(\frac{E_{0i}}{v_1} - \frac{E_{0r}}{v_1}\right)}_{=B_{1\parallel}} = \frac{1}{\mu_2}\underbrace{\frac{E_{0t}}{v_2}}_{=B_{2\parallel}} \Rightarrow E_{0i} - E_{0r} = \beta E_{0t}$$

Since the electromagnetic wave is incident normal to the interface, we have

$$B_{\perp} = 0$$
$$E_{\perp} = 0 \Rightarrow \sigma_f = 0$$

Assume $\vec{K}_f = 0$, we have

$$\frac{B_{1\parallel}}{\mu_1} - \frac{B_{2\parallel}}{\mu_2} = 0 \Rightarrow \frac{1}{\mu_1}\underbrace{\left(\frac{E_{0i}}{v_1} - \frac{E_{0r}}{v_1}\right)}_{=B_{1\parallel}} = \frac{1}{\mu_2}\underbrace{\frac{E_{0t}}{v_2}}_{=B_{2\parallel}} \Rightarrow E_{0i} - E_{0r} = \tilde{\beta}E_{0t}$$

where

$$\tilde{\beta} = \frac{\mu_1 v_1}{\mu_2 v_2}\tilde{k}_2$$

Thus, we obtain

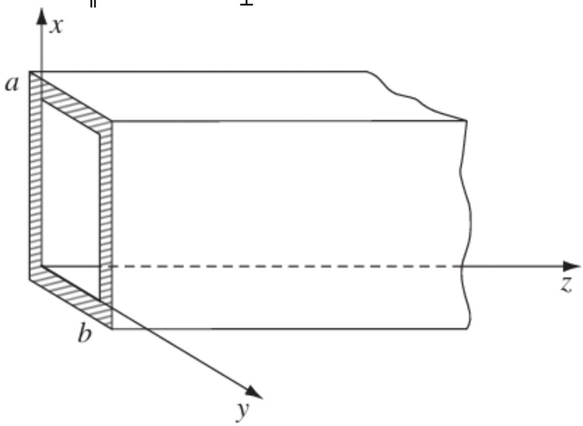
$$E_{0r} = \frac{1-\tilde{\beta}}{1+\tilde{\beta}}E_{0i}$$
$$E_{0t} = \frac{2}{1+\tilde{\beta}}E_{0i}$$

12-4 Wave Guides

A. HOLLOW WAVE GUIDE

(1) Consider a hollow pipe of rectangular shape. The electromagnetic wave is confined to the interior of the hollow pipe (rectangular wave guide). We assume that the wave guide is a perfect conductor, so that $\vec{E} = 0$ and $\vec{B} = 0$ inside the material, and hence the boundary conditions at the inner wall are

$$\vec{E}_{\parallel} = 0 \text{ and } \vec{B}_{\perp} = 0$$



In the interior of the wave guide where there is no free charges and currents, Maxwell's equations becomes

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \cdots\cdots \text{Faraday's law}$$
$$\nabla \cdot \vec{E} = 0 \cdots\cdots \text{Gauss's law}$$
$$\nabla \times \vec{B} = \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t} \cdots\cdots \text{Displacement current}$$
$$\nabla \cdot \vec{B} = 0$$

(2) Suppose

$$\vec{E}(x,y,z,t) = \vec{E}_0(x,y,z)e^{i(kz-\omega t)}$$
$$\vec{B}(x,y,z,t) = \vec{B}_0(x,y,z)e^{i(kz-\omega t)}$$

where

$$\vec{E}_0(x,y,z) = E_x\hat{x} + E_y\hat{y} + E_z\hat{z}$$
$$\vec{B}_0(x,y,z) = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}$$

and use Faraday's law and the displacement current. We obtain

$$\underbrace{\begin{aligned} &\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = i\omega B_z, \\ &\frac{\partial E_z}{\partial y} - ikE_y = i\omega B_x, \\ &ikE_x - \frac{\partial E_z}{\partial x} = i\omega B_y, \end{aligned}}_{\text{Faraday's law}} \qquad \underbrace{\begin{aligned} &\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = -\frac{i\omega}{c^2}E_z \\ &\frac{\partial B_z}{\partial y} - ikB_y = -\frac{i\omega}{c^2}E_x \\ &ikB_x - \frac{\partial B_z}{\partial x} = -\frac{i\omega}{c^2}E_y \end{aligned}}_{\text{Displacement current}}$$

If $E_z = 0$, we call these TE wave (transverse electric wave).
If $B_z = 0$, we call these TM wave (transverse magnetic wave).
If both $E_z = 0$ and $B_z = 0$, we call these TEM wave (transverse electromagnetic wave).

(3) From Gauss's law, if $E_z = 0$, we have

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} = 0 \Rightarrow \nabla \cdot \vec{E}_0 = 0$$

From Faraday's law, if $B_z = 0$, we have

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0 \Rightarrow \nabla \times \vec{E}_0 = 0$$

Thus, $\vec{E}_0$ can be written as

$$\vec{E}_0 = \nabla\varphi$$

satisfying Laplace's equation. Since $\vec{E}_{\parallel} = 0$ requires that the surface be an equipotential, and Laplace's equation has no local maxima and minima, thus the potential is constant throughout. Hence,

$$\vec{E}_0 = \nabla\varphi = 0$$

$\Rightarrow$ No TEM wave in a completely empty pipe.

(4) Consider the propagation of TE waves, i.e., $E_z = 0$

$$\left.\begin{aligned} E_z &= 0 \\ \frac{\partial E_z}{\partial y} - ikE_y &= i\omega B_x \\ ikE_x - \frac{\partial E_z}{\partial x} &= i\omega B_y \end{aligned}\right\} \Rightarrow E_y = -\frac{\omega}{k}B_x \text{ and } E_x = \frac{\omega}{k}B_y$$

$$\left.\begin{aligned}\frac{\partial E_y}{\partial x}-\frac{\partial E_x}{\partial y}&=i\omega B_z\\ \frac{\partial B_z}{\partial y}-ikB_y&=-\frac{i\omega}{c^2}E_x\\ ikB_x-\frac{\partial B_z}{\partial x}&=-\frac{i\omega}{c^2}E_y\end{aligned}\right\}\Rightarrow\left[\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\left(\frac{\omega}{c}\right)^2-k^2\right]B_z=0$$

Using separation of variables, we have

$$B_z(x,y)=\mathrm{X}(x)\mathrm{Y}(y)$$

so that

$$\underbrace{\frac{1}{\mathrm{X}}\frac{\partial^2}{\partial x^2}\mathrm{X}}_{=-k_x^2}+\underbrace{\frac{1}{\mathrm{Y}}\frac{\partial^2}{\partial y^2}\mathrm{Y}}_{=-k_y^2}+\left[\left(\frac{\omega}{c}\right)^2-k^2\right]=0$$

with $-k_x^2-k_y^2+\left[\left(\frac{\omega}{c}\right)^2-k^2\right]=0$

Thus, we then obtain two ordinary differential equations and solutions:

$$\frac{1}{\mathrm{X}}\frac{\partial^2}{\partial x^2}\mathrm{X}=-k_x^2\Rightarrow\mathrm{X}(x)=A\sin k_x x+B\cos k_x x$$
$$\frac{1}{\mathrm{Y}}\frac{\partial^2}{\partial y^2}\mathrm{Y}=-k_y^2\Rightarrow\mathrm{Y}(y)=C\sin k_y y+D\cos k_y y$$

Since

$$\mathrm{X}(0)=0\Rightarrow A=0\Rightarrow\mathrm{X}(x)=B\cos k_x x$$
$$\mathrm{X}(a)=0\Rightarrow\cos k_x a=0\Rightarrow k_x=\frac{m\pi}{a},\qquad m=0,1,2,\cdots$$
$$\mathrm{Y}(0)=0\Rightarrow C=0\Rightarrow Y(y)=D\cos k_y y$$
$$\mathrm{Y}(b)=0\Rightarrow\cos k_y b=0\Rightarrow k_y=\frac{n\pi}{b},\qquad m=0,1,2,\cdots$$

we obtain

$$B_z(x,y)=B_0\cos\frac{m\pi}{a}x\cos\frac{n\pi}{b}y$$

This solution is called the TE_{mn} mode.

(5) For TE_{00} mode, we have

$$B_z(x,y)=B_0=\text{constant}$$

Using Faraday's law, we have

$$\oint_{\mathcal{C}}\vec{E}\cdot d\vec{s}=-\frac{d}{dt}\int_{\mathcal{S}}\vec{B}\cdot d\vec{a}=i\omega B_z e^{i(kz-\omega t)}\int_{\mathcal{S}}da=iab\omega B_z e^{i(kz-\omega t)}$$

Since inside the metal, $\vec{E}=0$, thus we obtain

$$\oint_{\mathcal{C}}0\cdot d\vec{s}=0=iab\omega B_z e^{i(kz-\omega t)}\Rightarrow B_z=0$$

Therefore, TE_{00} mode is a TEM model. Thus, TE_{00} mode cannot occur in a rectangular wave guide.

(6) Since

$$-k_x^2 - k_y^2 + \left[\left(\frac{\omega}{c}\right)^2 - k^2\right] = 0$$

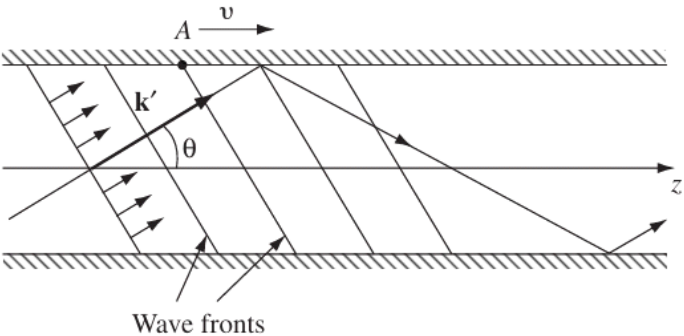
$$\Rightarrow k^2 = \left(\frac{\omega}{c}\right)^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2$$

Let

$$\omega_{mn} = c\pi\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\Rightarrow k = \frac{1}{c}\sqrt{\omega^2 - \omega_{mn}^2}$$

The propagation of an electromagnetic wave in a rectangular wave guide,



where

$$\vec{k}' = k_x\hat{x} + k_y\hat{y} + k_z\hat{z} = \frac{m\pi}{a}\hat{x} + \frac{n\pi}{b}\hat{y} + k\hat{z}$$

$$v = \frac{\omega}{k} = \frac{c}{\sqrt{1 - (\omega_{mn}/\omega)^2}}$$

B. THE COAXIAL TRANSMISSION LINE

(1) Consider a long straight wire of radius a surrounded by a cylindrical conducting sheath of radius b.

Since the interior of the long straight wire is not empty, TEM modes with $E_z = 0$ and $B_z = 0$ can occur.
Thus, Faraday's law and the displacement current give

$$\left.\begin{aligned} E_z &= 0 \\ \frac{\partial E_z}{\partial y} - ikE_y &= i\omega B_x \\ ikE_x - \frac{\partial E_z}{\partial x} &= i\omega B_y \end{aligned}\right\} \Rightarrow E_y = -\frac{\omega}{k}B_x \text{ and } E_x = \frac{\omega}{k}B_y$$

$$\left.\begin{aligned} B_z &= 0 \\ \frac{\partial B_z}{\partial y} - ikB_y &= -\frac{i\omega}{c^2}E_x \\ ikB_x - \frac{\partial B_z}{\partial x} &= -\frac{i\omega}{c^2}E_y \end{aligned}\right\} \Rightarrow B_y = \frac{\omega}{kc^2}E_x \text{ and } B_x = -\frac{\omega}{kc^2}E_y$$

We obtain

$$E_x = \frac{\omega}{k}B_y = \frac{\omega}{k}\frac{\omega}{kc^2}E_x \Rightarrow c = \frac{\omega}{k}$$

so the waves travel at speed c and are non-dipersive.
Since

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0 \Rightarrow \nabla \times \vec{E} = 0 \text{ and } \nabla \cdot \vec{E} = 0$$

$$\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = 0 \Rightarrow \nabla \times \vec{B} = 0 \text{ and } \nabla \cdot \vec{B} = 0$$

These are the equations of electrostatics and magnetostatics, for empty space in 2D. The solutions are

$$\vec{E}_0(r,\phi) = \frac{A}{r}\hat{r}, \qquad \vec{B}_0(r,\phi) = \frac{A}{cr}\hat{\phi}$$